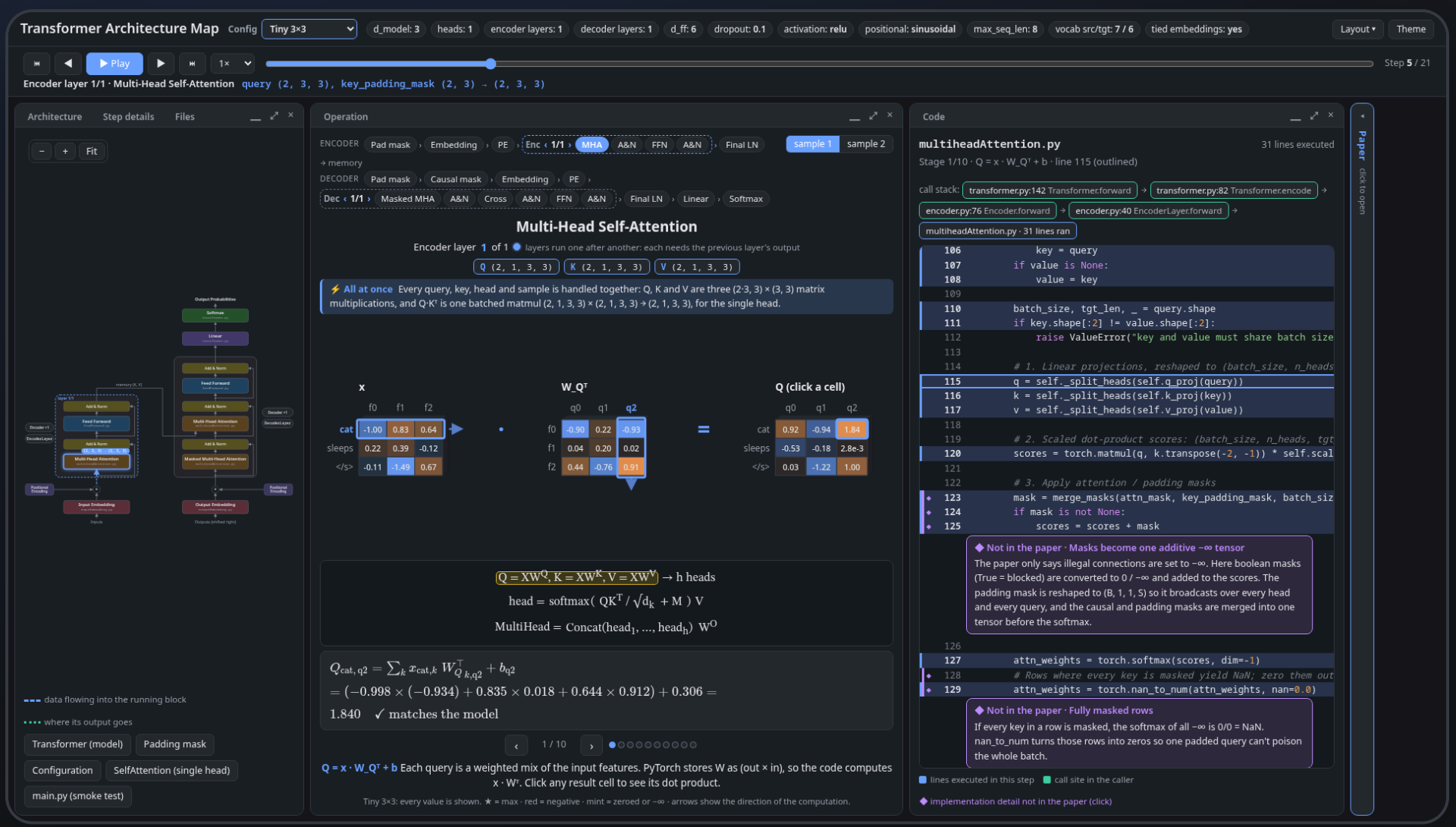
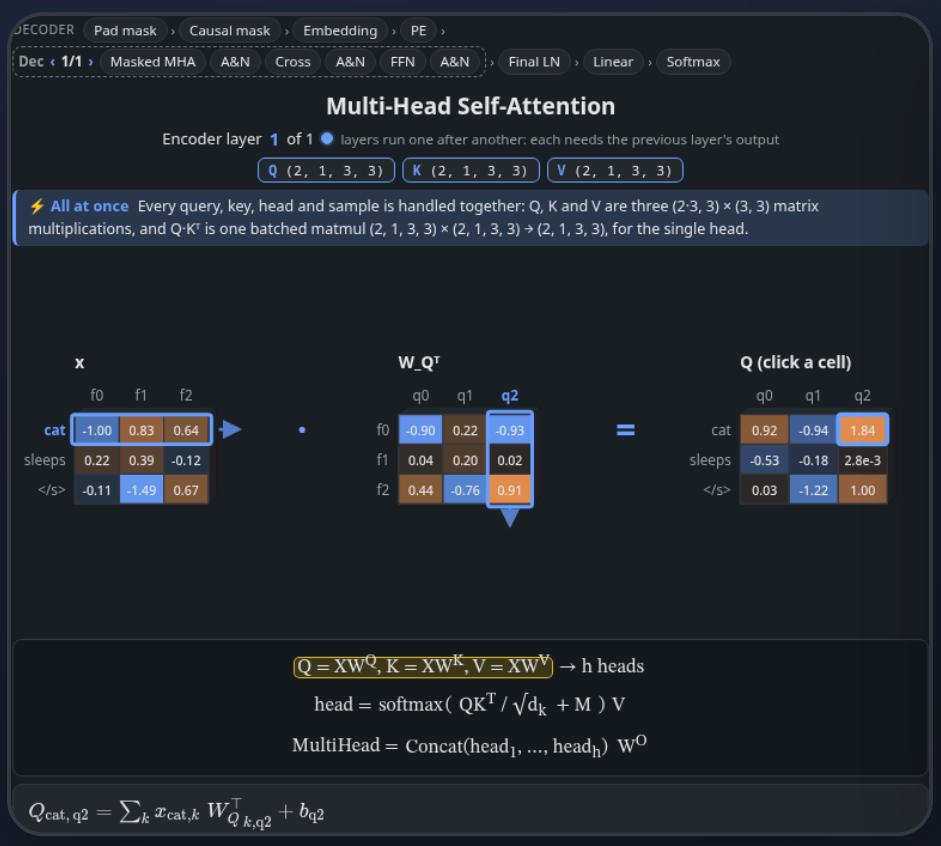
See every number inside a Transformer

An interactive walkthrough of a real PyTorch encoder–decoder forward pass: one tensor operation at a time, linked to the exact line of code that runs it.



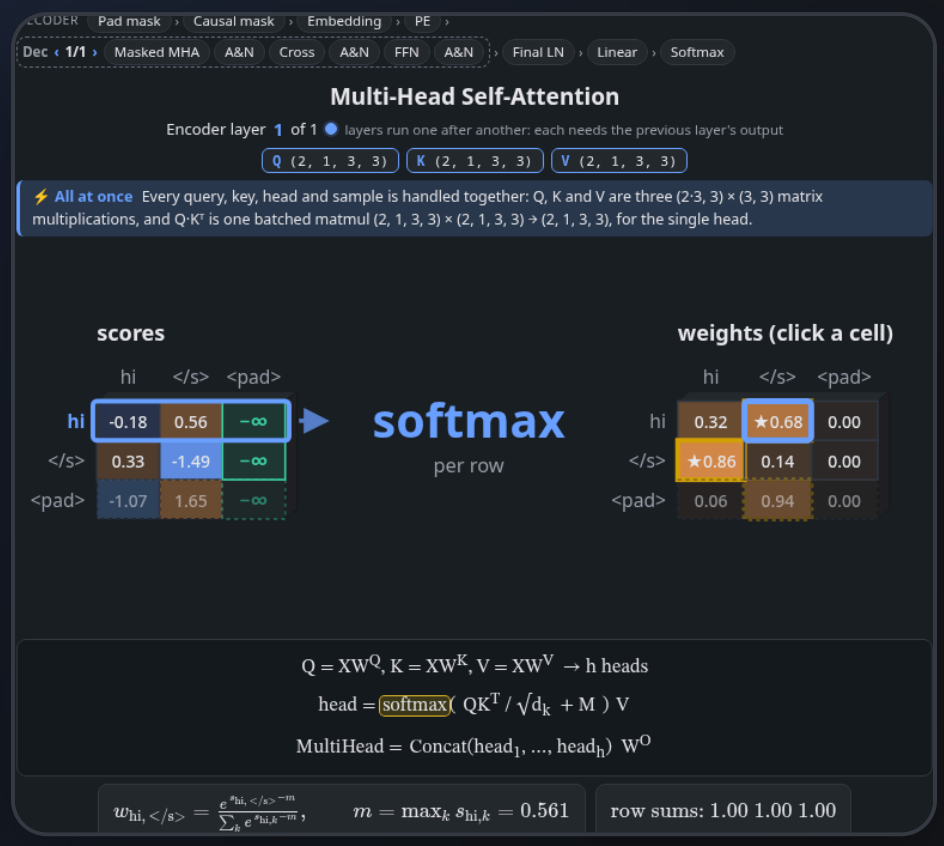
Swipe →

Every step, worked out



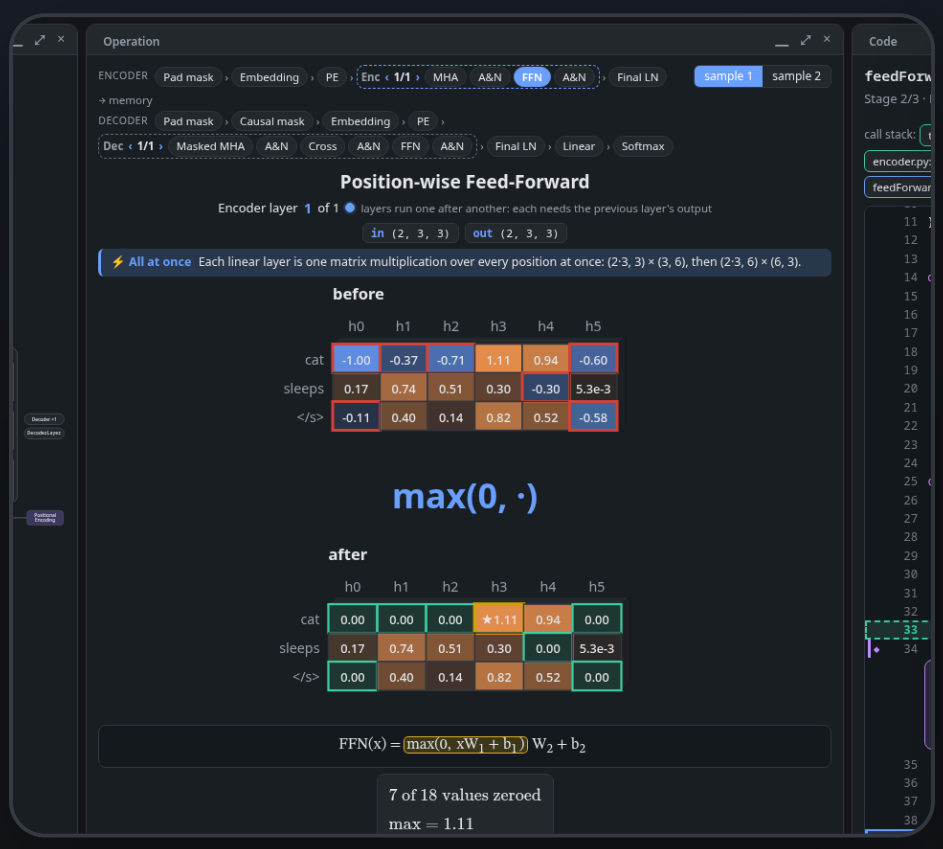
Matmul, cell by cell

Click any result: its row × column lights up and the dot product is checked against the model.



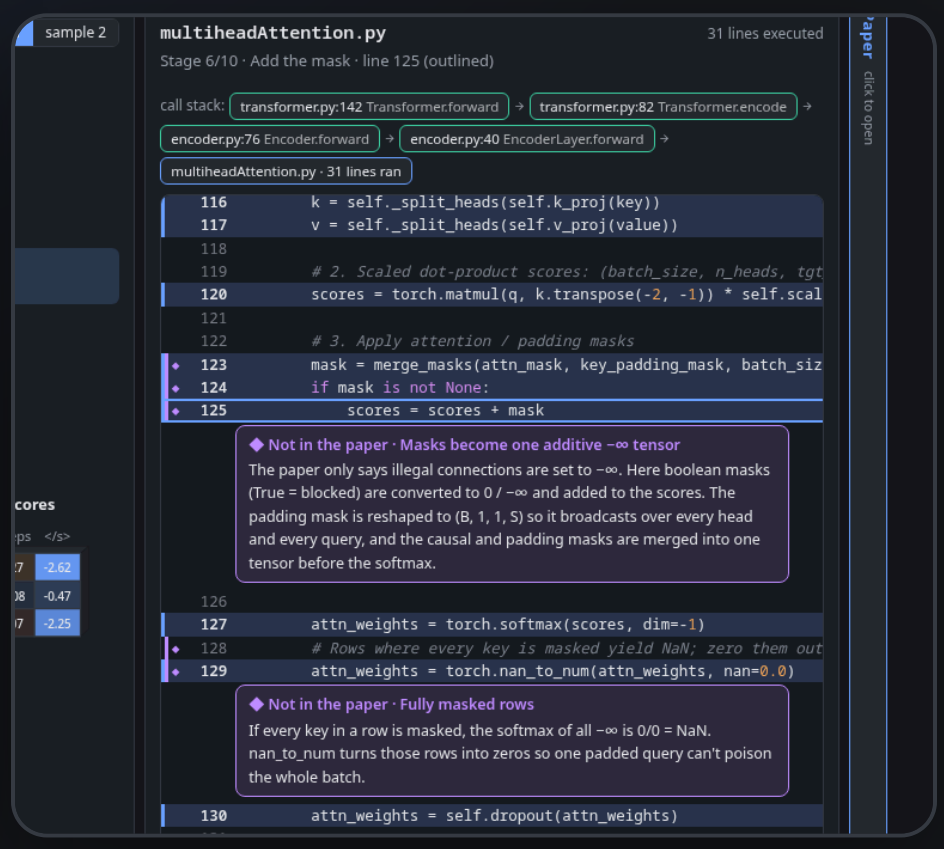
LaTeX arithmetic

Softmax, LayerNorm and positional encoding written out with the real values.



Special values flagged

ReLU zeros, the row max, −∞ masks and padding, highlighted where they happen.

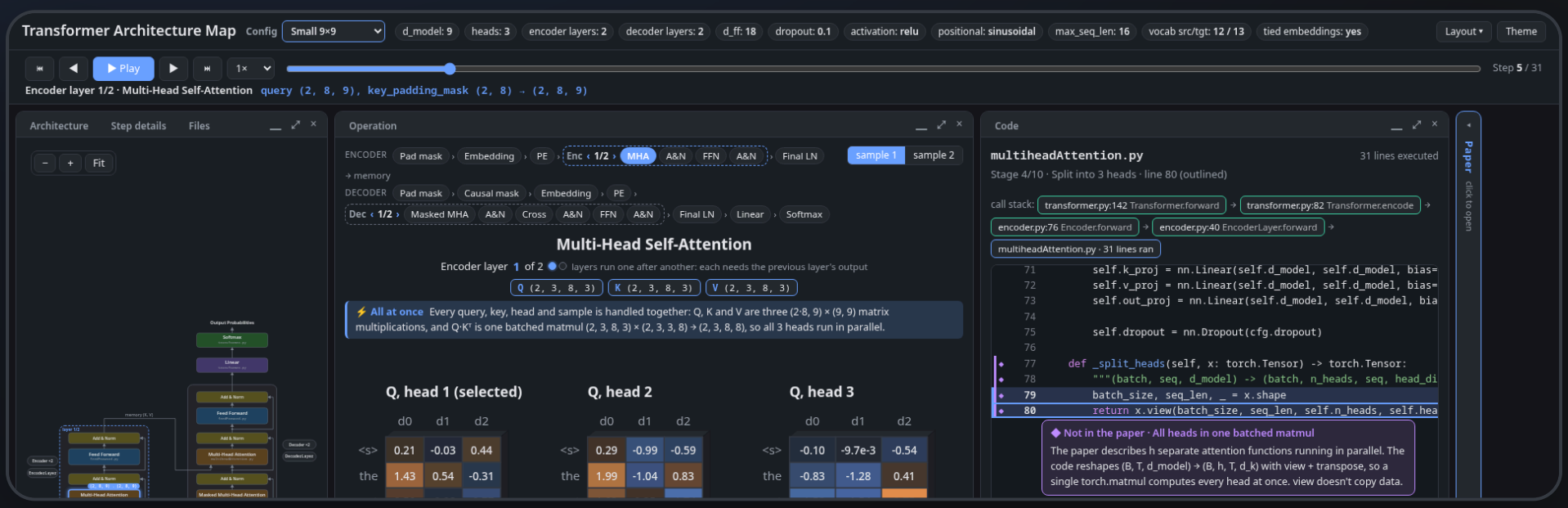


Code in sync

Each stage outlines the line that runs it, plus implementation details not in the paper.

Small enough to check by hand

- **Three model sizes:** Tiny 3×3 (every value on screen), Small 9×9 (3 heads, 2 + 2 layers), and the full config.
- **Real execution trace:** captured from the PyTorch model with hooks and sys.settrace; nothing is hand-drawn.
- **Isometric tensors** with the batch, sequence and feature axes named, plus "all at once" notes on parallelism.
- **Dockable workspace:** drag, dock and collapse panels; architecture diagram, code and paper notes stay linked.



`github.com/omarTBakr/TransformersVisualizations`

Open source. Next: decoder-only (GPT-style) and encoder-only (BERT-style).